

VANCOUVER HARBOUR, BC, Canada

WMO: 712010

Lat: 49.295N

Lon: 123.122W

Elev: 8

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.2	30.5	9.9	9.1	32.3	16.7	12.8	34.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	78.8	64.6	76.2	63.6	73.8	62.5	66.2	76.1	64.9	74.4	63.7	72.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	82.7	71.1	60.7	79.4	69.4	59.7	76.4	68.1	30.8	76.5	29.9	74.5	29.0	72.8	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	23.9	84.6	4.2	3.0	20.8	86.8	18.3	88.5	16.0	90.2	12.9	92.4
			WB	20.4	68.3	3.5	1.6	17.9	69.5	15.8	70.4	13.8	71.3	11.2	72.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	41.4	42.2	45.4	50.3	56.7	61.4	66.0	65.8	60.1	51.8	45.2	41.0	(6)
	DBStd	10.02	5.24	4.47	4.13	4.04	4.51	4.19	3.74	3.31	3.73	3.74	4.94	5.11	(7)
	HDD50	1140	268	219	150	43	3	0	0	0	0	25	153	279	(8)
	HDD65	4762	732	639	607	441	257	125	32	30	152	410	593	744	(9)
	CDD50	1996	1	1	8	53	213	341	495	489	303	80	11	1	(10)
	CDD65	137	0	0	0	0	1	15	62	54	5	0	0	0	(11)
	CDH74	500	0	0	0	0	9	70	225	177	19	0	0	0	(12)
	CDH80	48	0	0	0	0	0	5	31	11	1	0	0	0	(13)
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15) Precipitation	PrecAvg	88.3	12.5	8.1	8.5	6.0	4.1	3.4	2.1	2.8	4.2	10.0	13.7	12.8	(15)
	PrecMax	111.6	20.8	16.1	15.8	12.4	6.8	6.6	5.0	12.6	9.2	17.1	25.7	18.5	(16)
	PrecMin	70.1	7.0	0.4	2.0	2.2	1.0	1.0	0.5	0.4	0.6	1.6	6.2	7.9	(17)
	PrecStd	10.0	3.3	3.2	2.8	2.3	1.4	1.6	1.2	2.6	2.3	3.7	4.8	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	53.7	59.3	67.3	74.9	80.1	83.5	81.4	76.5	64.5	57.2	53.5	(19)
		MCWB	49.1	49.7	49.4	54.9	59.5	62.9	66.0	64.9	63.2	56.4	53.7	50.2	(20)
	2%	DB	50.7	51.1	55.5	62.8	71.1	75.7	79.2	78.2	72.6	60.9	54.8	50.5	(21)
		MCWB	47.4	46.8	48.2	53.2	58.6	63.0	64.9	64.7	61.6	54.3	51.9	47.3	(22)
	5%	DB	49.3	49.4	53.1	59.7	67.7	72.2	76.1	75.8	69.5	58.7	53.1	48.7	(23)
		MCWB	46.4	45.9	46.9	51.0	57.4	61.2	63.7	64.0	59.9	53.8	50.5	46.0	(24)
	10%	DB	48.0	48.0	51.2	57.0	64.7	69.3	73.8	73.3	66.8	57.1	51.6	47.4	(25)
		MCWB	45.4	44.9	46.1	49.4	55.9	59.5	62.7	62.8	58.8	52.9	49.3	45.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	50.4	52.2	56.6	62.0	66.1	69.0	67.5	65.3	59.1	54.8	51.2	(27)
		MCDB	51.1	52.1	57.8	64.5	71.7	76.5	82.8	78.9	73.7	62.9	56.3	53.2	(28)
	2%	WB	48.1	48.4	49.9	53.9	60.1	64.0	66.4	66.1	62.9	56.7	52.3	48.0	(29)
		MCDB	49.5	50.3	54.4	60.6	69.5	73.5	76.6	76.0	70.2	59.5	54.1	49.6	(30)
	5%	WB	46.6	46.8	48.4	52.0	58.5	61.8	64.8	65.0	61.2	54.9	50.9	46.5	(31)
		MCDB	48.4	48.9	52.2	58.1	66.6	71.0	74.7	74.3	67.1	57.2	52.6	48.3	(32)
	10%	WB	45.2	45.4	47.0	50.3	57.1	60.0	63.6	63.8	59.7	53.4	49.5	45.0	(33)
		MCDB	47.2	47.7	50.5	54.9	64.2	68.1	72.9	72.2	64.7	56.3	51.0	46.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	8.0	9.1	11.2	12.4	12.2	13.6	13.4	11.6	8.3	6.9	6.1	(35)
		MCDBR	6.9	9.2	11.8	15.7	17.8	17.7	17.8	17.9	16.4	10.6	7.3	6.5	(36)
		MCWBR	5.5	6.1	7.4	8.3	8.8	7.9	7.7	7.6	7.7	6.4	5.8	4.8	(37)
	5% WB	MCDBR	6.4	7.2	11.1	13.2	15.9	16.3	17.1	16.7	13.7	8.7	6.7	5.9	(38)
		MCWBR	5.3	5.3	7.2	7.7	8.3	7.9	7.6	7.5	7.1	5.7	5.5	4.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.319	0.313	0.331	0.366	0.391	0.367	0.364	0.368	0.350	0.339	0.323	0.307	(40)
	taud		2.405	2.472	2.443	2.367	2.340	2.434	2.465	2.473	2.523	2.534	2.487	2.429	(41)
	Ebn at Noon		232	267	280	279	276	283	282	276	271	254	230	220	(42)
	Edh at Noon		21	24	30	35	38	35	33	32	27	23	19	18	(43)
(44) All-Sky Solar Radiation	RadAvg		269	541	806	1259	1653	1769	1984	1685	1163	638	316	214	(44)
	RadStd		46	83	114	139	145	217	180	135	106	91	61	34	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+32
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page